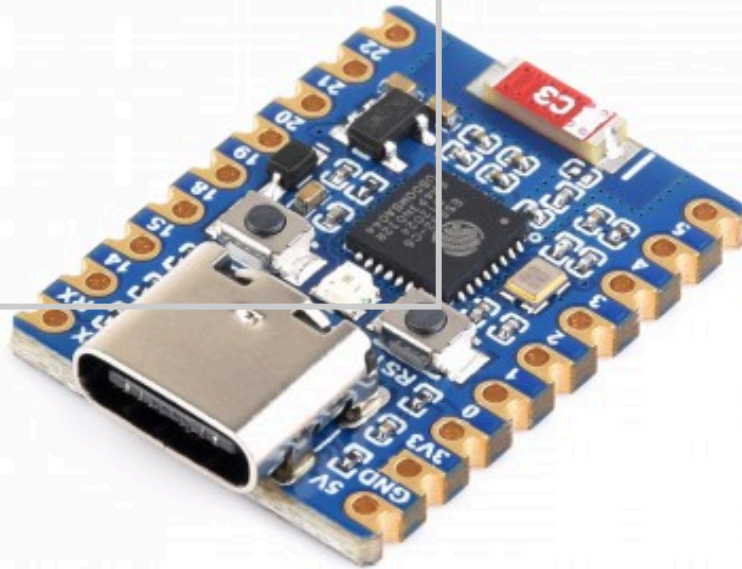
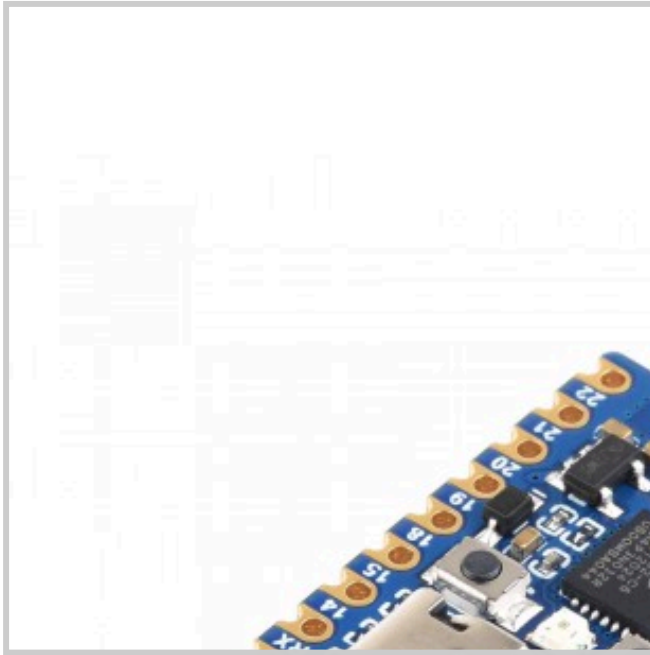




ESP32-C6

ESP32-C6 Mini Development Board, Based on ESP32-C6FH8, Dual Processors, 160MHz Running Frequency, 2.4GHz WiFi 6 & Bluetooth 5, ESP32 Development Board

~~\$4.⁹⁹~~ ~~\$5.⁹⁹~~

ADD TO CART

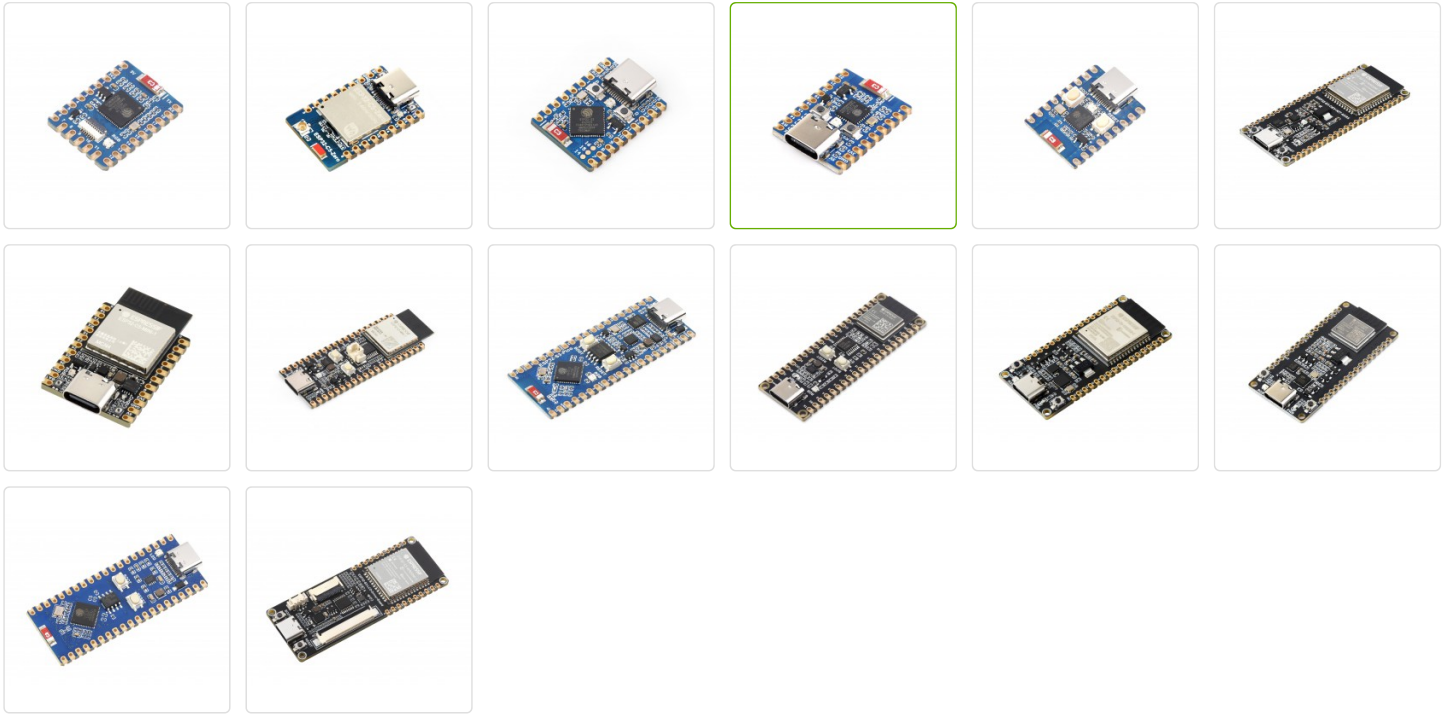
Version Options

with pre-soldered header

without header

Related Products:

Chat



[Description](#) [Package Content](#) [Wiki](#)

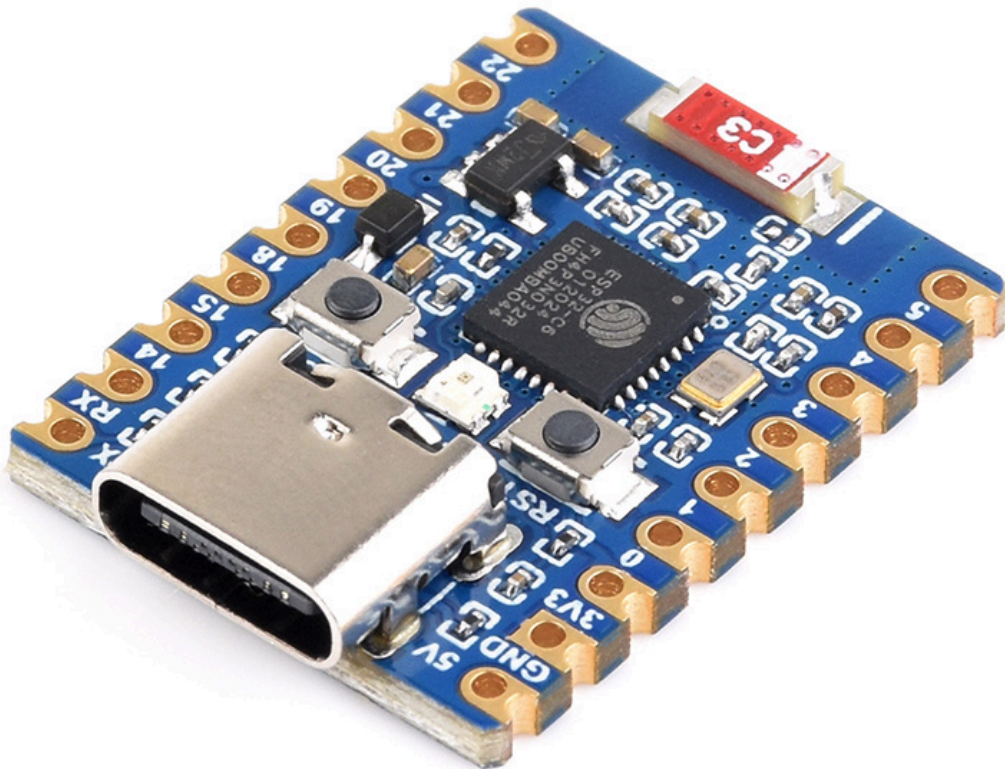
ESP32-C6 Mini Development Board, Based On ESP32-C6FH8, Dual Processors, 160MHz Running Frequency, 2.4GHz WiFi 6 & Bluetooth 5, ESP32 Development Board

ESP32-C6-Zero WiFi 6 Development Board

Based on ESP32-C6FH8, onboard multi-function GPIO pins

Castellated Module, Suitable For SMD Applications

[Chat](#)

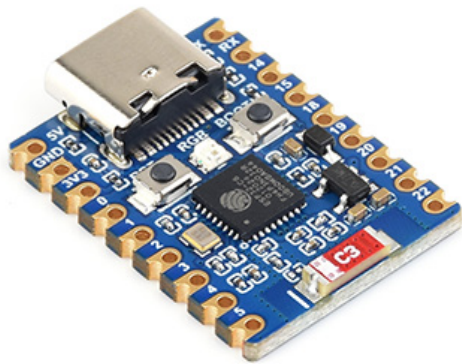


Board Specifications

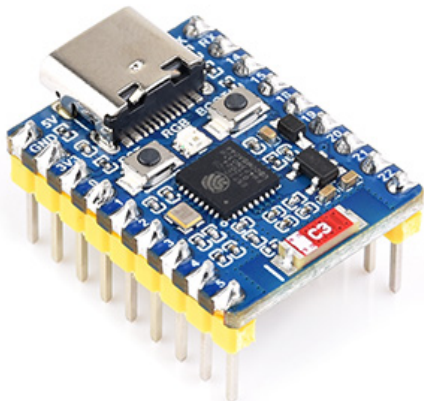
- Equipped with a high-performance 32-bit RISC-V processor with clock speed up to 160 MHz, and a low-power 32-bit RISC-V processor with clock speed up to 20MHz
- Built in 320KB ROM, 512KB of HP Static RAM, 16KB LP Static RAM and 8MB Flash memory

- Integrated WiFi 6, Bluetooth 5 and IEEE 802.15.4 (Zigbee 3.0 and Thread) wireless communication, with superior RF performance
- Castellated module and onboard ceramic antenna, allows soldering directly to carrier boards
- Supports flexible clock, module power supply independent setting, and other controls to realize low power consumption in different scenarios
- Onboard USB Type-C port, 22 × GPIO pins allows flexibly configuring pin functions

Version Options



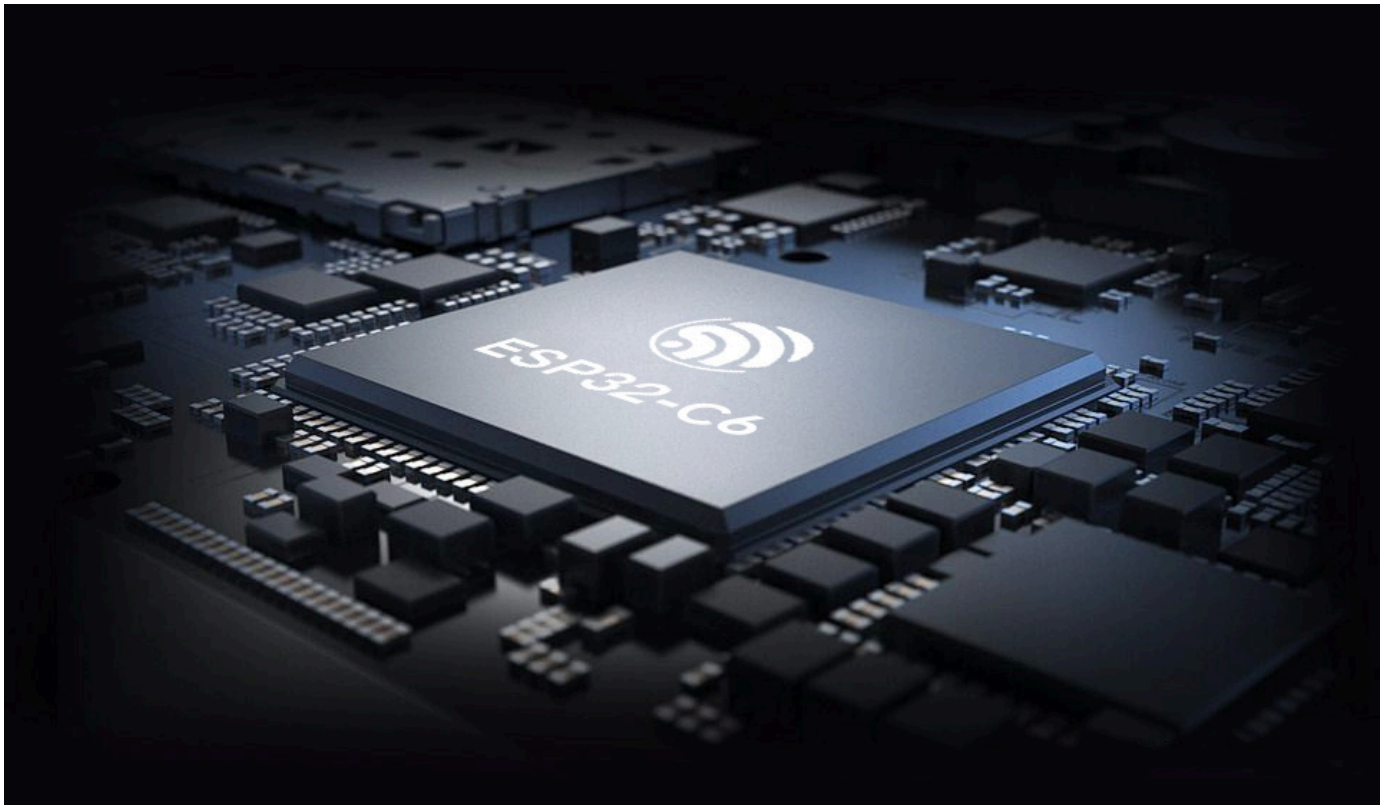
ESP32-C6-Zero
without pinheader



ESP32-C6-Zero-M
with pre-soldered header

ESP32-C6 Suitable For AIoT Applications

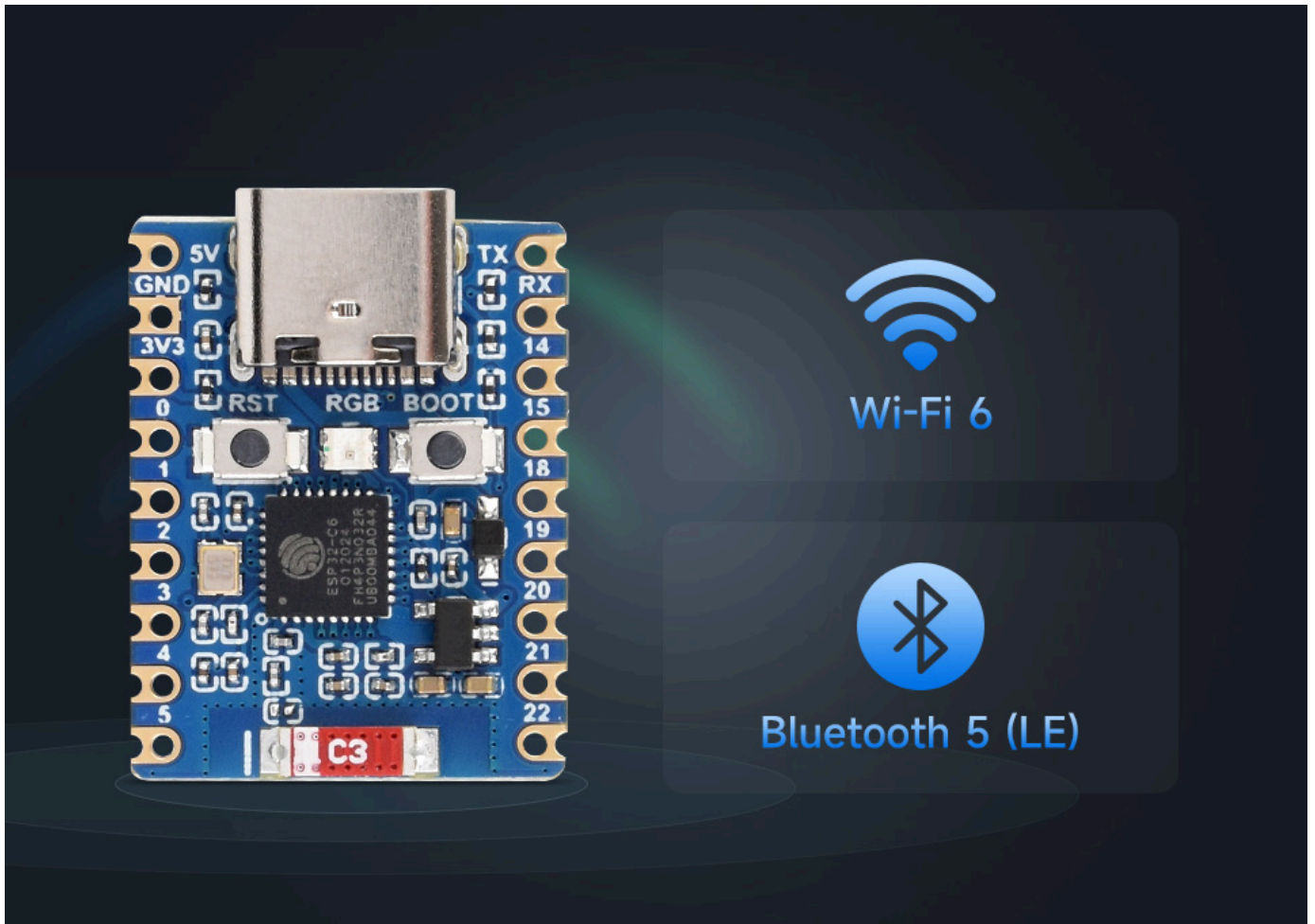
Powered By A High-Performance 32-Bit RISC-V Processor And A Low-Power 32-Bit RISC-V Processor, The Main Frequency Is Up To 160 MHz, Powerful AI Computing Capability & Reliable Security Features



WIFI 6 And BLE 5 Support

Supports WiFi 6, Bluetooth 5 And IEEE 802.15.4 (Zigbee 3.0 And Thread)

Its Bluetooth Low Energy Subsystem Supports Bluetooth 5 (LE) And Bluetooth Mesh



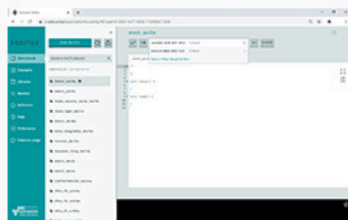
Supports ESP-IDF, Arduino

Comprehensive SDK, Dev Resources, Tutorials To Help You Easily Get Started



ESP-IDF

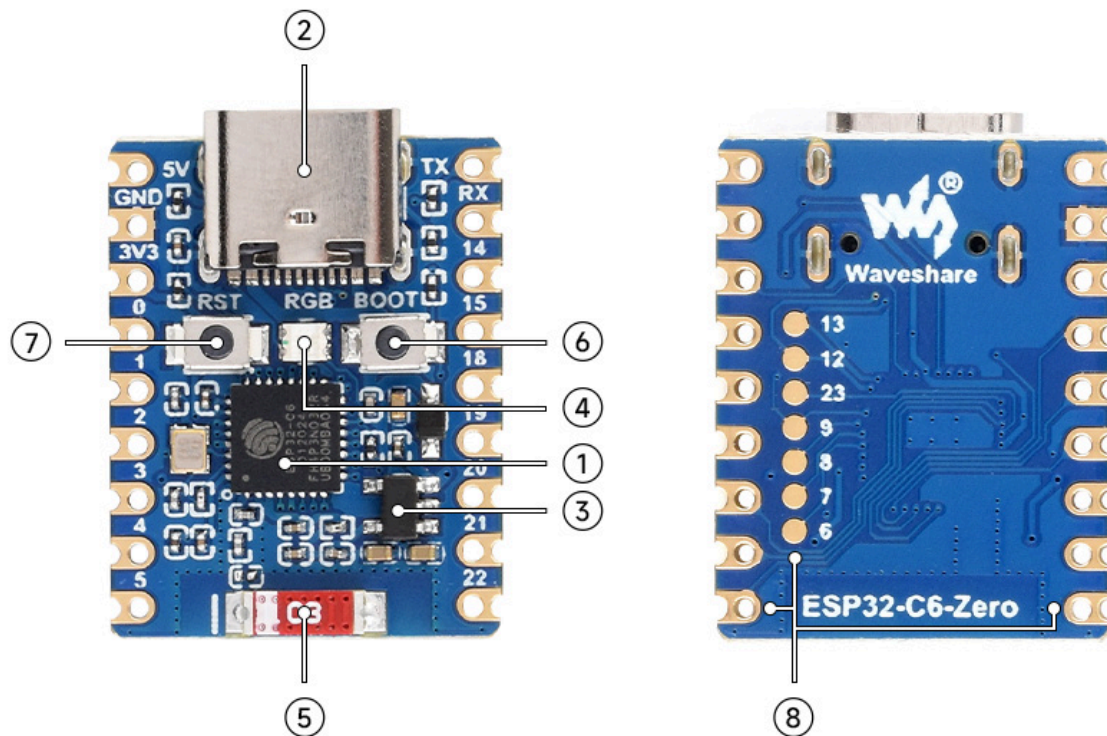
With free open source development tools, supports IDEs such as Eclipse and VSCode, easier for developers to use.



Arduino IDE

Arduino IDE is a open source electronic prototyping platform, convenient and flexible, easy to get started. After a simple learning, you can start to develop quickly.

What's On Board



1. ESP32-C6FH8

Built-in dual processors, main frequency is up to 160 MHz

2. USB Type-C Port

for downloading programme and debugging

3. ME6217C33M5G

low dropout LDO, 800mA (Max)

4. WS2812 RGB LED

5. 2.4G ceramic antenna

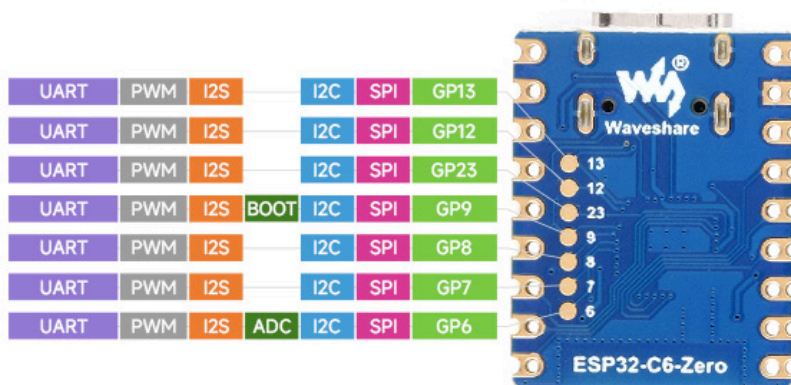
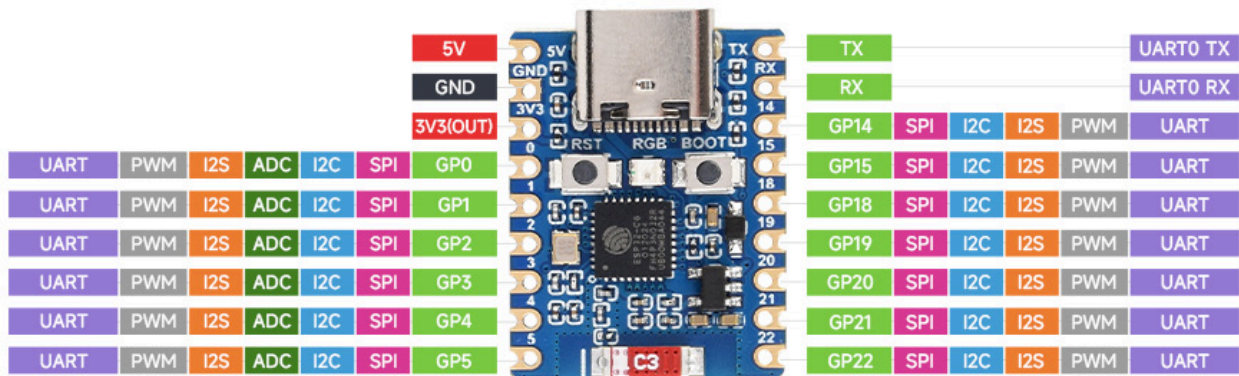
6. BOOT button

Press it and then press the RESET button to enter download mode

7. RESET button

8. ESP32-C6FH8 pins

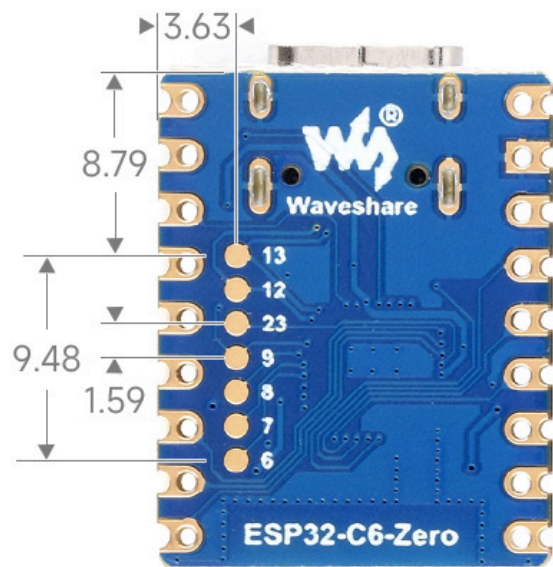
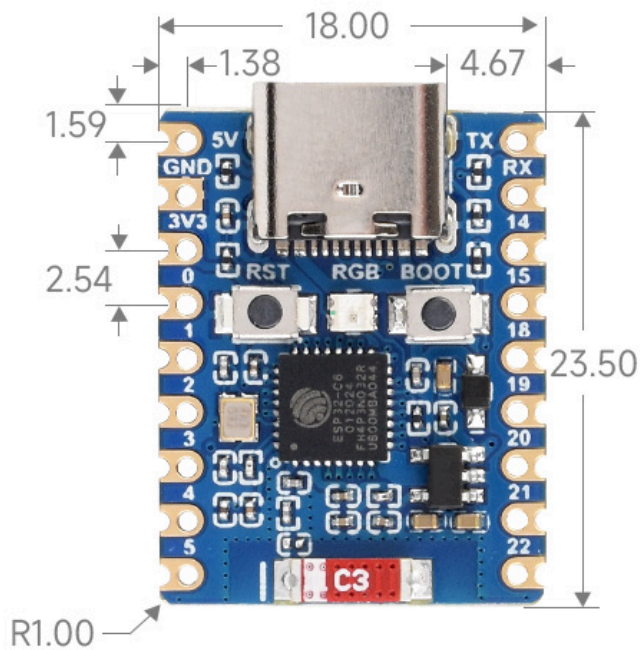
Pin Definition



WS2812 RGB LED used pin

GP8 DIN

Outline Dimensions



Unit:mm

Resources & Services



WAVESHARE
ORIGINAL ✓



PROFESSIONAL
TECH SUPPORT



RICH RESOURCES
MANUAL & DEMO

Wiki: docs.waveshare.com/ESP32-C6-Zero

ESP32-Zero Product Selection

PRODUCT	MCU	MAIN FREQUENCY	FLASH	SRAM	PSRAM	FEATURES

 ESP32-S3-Zero-N8R8	ESP32-S3-PICO-1-N8R8	240MHz	8MB	512KB	8MB	Supports 2.4GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE)
 ESP32-S3-Zero	ESP32-S3FH4R2	240MHz	4MB	512KB	2MB	Supports 2.4GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE)
 ESP32-C6-Zero	ESP32-C6FH8	160MHz	8MB	512KB	-	Supports Wi-Fi 6, Bluetooth 5 (LE), and IEEE 802.15.4 (Zigbee 3.0 and Thread)
 ESP32-C6-Zero-B	ESP32-C6FH8	160MHz	8MB	512KB	-	Supports Wi-Fi 6, Bluetooth 5 (LE), IEEE 802.15.4 (Zigbee 3.0 and Thread), and 5~36V power input
 ESP32-C5-Zero	ESP32-C5-HF4	240MHz	4MB	384KB	-	Supports 2.4GHz and 5GHz dual-band Wi-Fi, Bluetooth 5 (LE), and IEEE 802.15.4 (Zigbee 3.0 and Thread)
 ESP32-C3-Zero	ESP32-C3FH4	160MHz	4MB	400KB	-	Supports 2.4GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE)



[ESP32-H2-Zero](#)

ESP32-H2FH4S

96MHz

4MB

320KB

-

Supports Bluetooth 5 (LE) and IEEE 802.15.4 (Zigbee 3.0 and Thread)